

List of minimum Tools & Equipments for a batch of 30 students			
Electrical Engineering			
SECTOR : INFRASTRUCTURE			
NSQF LEVEL - 4			
Sl No	Name of the Tool	Specification	Quantity
1	Wire Strippers	Adjustable	20
2	Sheath Stripper	Adjustable	20
3	Wire stripper pliers	Adjustable	20
4	Insulated Screw driver	Flat or Star inter changeable	20
5	Screw driver set	Multi bit set of 6	20
6	Screw driver	Flat standard 3 mm tip long reach	20
7	Screw driver	Flat standard 8 mm tip long reach	20
8	Screw driver	Flat standard 10 mm tip long reach	20
9	Screw driver	Flat standard 10 mm tip 250mm	20
10	Pliers	Insulated long nose	20
11	Pliers	Insulated combination plier	20
12	Pliers	Side cutting plier	20
13	Pliers	Circlip remover straight external	20
14	Pliers	Bent nose circlip	20
15	Pliers	End cutters 180 mm	20
16	Hacksaws	Adjustable 300mm	20
17	Mallet	Wooden	20
18	Cable cutters	Ratchet cable cutter 250 mm	20
19	Hammer	Ball Peen 200 gm	20
20	Hammer	Claw Hammer Steel 500 gm	20
21	Adjustable Spanner	34 mm Jaw capacity, length 300 mm	10
22	Adjustable Spanner	20 mm Jaw capacity	10
23	Spanner Sets	Open end	10
24	Spanner Sets	Ring Spanner	5
25	Box Spanner	Socket Set	2
26	Electricians' Safety Knife	Retractable Heavy duty with straight blade	20
27	Hex Keys	T handle set	5
28	Line tester	500 v	20
29	Multimeter	Analog Hand held type	10
30	Multimeter	DC 200mv -1000v, 0 – 10A & AC 200mv- 750v , 0-10A, resistance 0-20 MΩ and 3 1/2 digit	10
31	Tong Tester	400 Amps	5
32	MULTI TESTER	PF/KVA/KW/FREQUENCY Integrated Instrument	2

Sl No	Name of the Tool	Specification	Quantity
33	Hand gloves	14" - Leather	20
34	Hand gloves	8" Cotton	20
35	Goggles		5
36	Ohm Meter; Series Type & Shunt Type, portable box type	50/2000-ohm analog	5
37	A.C. Voltmeter M.I. analog, portable box type housed in Bakelite case	Multi range 75 V - 150V - 300V - 600V	10
38	Milli Voltmeter center zero analog, portable box type housed in Bakelite case	100 – 0 – 100 mV	10
39	Ammeter MC analog, portable box type housed in Bakelite case	0 - 500 mA, 0-5 A, 0-25 A	10
40	AC Ammeter MI, analog, portable box type housed in Bakelite case	0 - 1 A, 0-5 A, 0-25 A	10
41	Kilo Wattmeter Analog	0-1.5-3KW, pressure coil rating- 240v/440v, current rating-5A/10A Analog, portable type Housed in Bakelite case	10
42	Digital Wattmeter	230 V, 1 KW, 50 Hz	10
43	A.C. Energy Meter	Single Phase, 10 A, 240 V induction type (as per IEC 61850)	5
44	A.C. Energy Meter	Three Phase, 15 A , 440 V induction type (as per IEC 61850)	3
45	Digital Energy Meter	Single Phase, three phase (as per IEC 61850)	3
46	Power Factor Meter Digital	440 V, 20 A, Three Phase portable box type	2
47	Frequency Meter	45 to 55 Hz	2
48	Magnetic Flux Meter	0-500 Tesla	2
49	Lux meter	Lux meter LCD read out 0.05 to 7000 lumens with battery.	2
50	Tachometer	Analog Type - 10000 RPM	2
51	Tachometer	Digital Photo Sensor Type - 10000 RPM	2
53	Hand Drill Machine	0-6 mm capacity	1
54	Portable Electric Drill Machine	0-12 mm capacity 750w, 240v with chuck and key	2
55	Load Bank (Lamp / heater Type)	6 KW, 3Ph	3
56	Brake Test arrangement with two spring balance rating	0 to 25 kg	1
57	Tong Tester / Clamp Meter	0 - 100 A (Digital Type)	3
58	Megger	Analog - 500 V	2
59	Earth Resistivity tester		1
60	Wheat Stone Bridge with galvanometer and battery		3
61	Single Phase Variable Auto Transformer	0 - 270 V, 10Amp (Air cooled)	3
62	Phase Sequence Indicator	3 Phase, 415 V	2

Sl No	Name of the Tool	Specification	Quantity
63	AC Starters: - a. Resistance type starter b. Direct on line Starter c. Star Delta Starter- Manual d. Star Delta Starter – Semi automatic e. Star Delta Starter – Fully automatic f. Star Delta Starter - Soft starter	For A.C Motors of 2 to 5 H.P.	2
64	Oscilloscope Dual Trace	20 MHz	2
65	Synchroscope	440V, 50 Hz	2
66	Function Generator	2 to 200 KHz, Sine, Square, Triangular 220 V, 50 Hz, Single Phase	2
67	Digital multi-function meter	3 Phase	2
68	Soldering Iron	25-Watt, 65 Watt and 120-Watt, 230 Volt	10
69	Temperature controlled Soldering Iron	50-Watt, 230 Volt	2
70	Oil Testing Kit	Oil Testing Kit 230 V, single phase 50 Hz 60 VA output 0-60 KV Variable	1
71	Inverter with Battery	1 KVA with 12 V Battery Input- 12-volt DC Output- 220 volt AC	2
72	Ni-Cd Battery	1.2 Amps	4
73	Voltage Stabilizer	AC Input - 150 - 250 V, 600 VA AC Output - 240 V, 10 A	2
74	DC Power Supply	0 - 30 V, 5 A	6
75	24 V battery set		3
76	110 V battery charger		1
77	Battery Charger	0 - 6 - 9 - 12 - 24 - 48 V, 30amp	1
78	Current Transformer	415 V, 50Hz, CT Ratio 25 / 5 A, 5VA	3
79	Potential Transformer	415 V, 50Hz, PT Ratio, 440V/110V, 10VA	3
80	Solar panel with Battery	18 Watt	2
81	D.C. milli ammeter	0-500m A	5
82	Potential Transformer	415 volt, 50 Hz, PT ratio 11KV/ 110 V, 10VA	1

Common facility for all programmes in an Institution			
CLASS ROOM FURNITURE			
1	Instructors table		1 no.
2	Instructors chair		1 no.
3	Students chair with writing pads/bench-desk arrangement		for 30 students
4	Almirah		1 no
5	Black/White board	1200 x 900 mm	1 no.
6	Instructors Laptop with internet facility	Windows with MS office package	1 no.
7	LCD Projector		1 no
IT LAB *			
1	Table & Chair for Instructors		Sufficient
3	Students chair with writing pads/bench-desk arrangement		for 30 students
4	Steel almarah / cup board		1 no.
5	Black/White board	1200 x 900 mm	1 no.
6	Desktop with internet facility	Windows with MS office package (Latest configuration)	10 nos.
7	LCD Projector		1 no
8	Printer	A4 Size	1 no.
9	Computer table		Sufficient number for arranging 10 computers

* Common for up to 5 programmes or 5 batches in an Institution. For Institutions with more than 5 programmes or 5 batches there shall be proportionate increase in number of computers and related furniture.